

PROVE THE PHOTO IS REAL · TAKEN NOW · OF THE UNIT IT CLAIMS TO BE

Six Pillars to make AutoKon's **evidence** stand up to **bank scrutiny.**

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BUILDS ON

Dickson evidence integrity
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FOR

Dickson · Faishal · BSN/BTN
pitch

DATE

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Banks need defensible proof. The current pipeline has a single trust surface.

TODAY · STRUCTURAL FRAUD GAPS

- Photo recycling and metadata manipulation
- Adjacent-unit confusion (FLPP 6m grid)
- Single Claude read = entire trust surface
- No cross-checking layer
- Indonesian field reality breaks naive GPS

WHAT BANKS ASK · 6 QUESTIONS

- Is this photo real? (Authenticity)
- Is the submitter who they claim? (Identity)
- Is this the actual unit? (Place)
- Was this captured now? (Time)
- Is the audit trail tamper-proof? (Custody)
- Do you trust other than your own software? (Independent)

Six pillars of trust. One bank question per pillar.

Reorganized from Dickson's 10-layer engineering architecture into 6 audience-first pillars that map directly to bank concerns. 35 items across all pillars; 100+ proposed solutions.

PILLAR 01 · 6 ITEMS

Authenticity

Is the photo real?

PILLAR 02 · 5 ITEMS

Identity

Who submitted it?

PILLAR 03 · 7 ITEMS

Place

Is this the right unit?

PILLAR 04 · 6 ITEMS

Time

Was it captured now?

PILLAR 05 · 6 ITEMS

Chain of Custody

Tamper-proof?

PILLAR 06 · 5 ITEMS

Independent Verification

External cross-check?

Authenticity

"Is the photo real, or is it fabricated, edited, or AI-generated?"

1.1	Photo recycling detection	pHash + deep-learning embedding · Dickson's L3 baseline cross-project layer	SHIPPED
1.2	Cross-photo consistency	Time-spread + shadow direction + Claude batch-check	
1.3	Edit/Photoshop forgery (ELA)	ELA + multi-model AI ensemble — 'as an AI tech company we should enable this'	
1.4	AI-generated photo detection	JAKI Pemprov DKI Gemini scandal proof point — must address	
1.5	Trusted-source overlay validation	Whitelisted timestamp camera + custom AutoKon app (PI commitment)	SHIPPED
1.6	Low-tech consumer-app fraud detection	Word export · IG Story · screenshots · photo-of-print · high leverage NEW	NEW

Identity

"Is the person submitting this who they claim to be?"

2.1	Phone-number binding	Phone whitelist · 2FA OTP · WhatsApp Business API · periodic auth	SHIPPED
2.2	KTP-bound capture	Pak Jun #10 · per-session KTP holding photo + KYC partnership (Privy/Tilaka)	
2.3	Voice-print of Pelaksana	Voice OTP at session start · piggybacks on L0b audio fingerprint	NEW
2.4	Liveness check	Passive + active liveness · WhatsApp-native KYC enrollment (Adrian's idea)	NEW
2.5	Device attestation	Android Play Integrity · pairs with custom AutoKon camera	NEW

Place

"Is this the actual unit it claims to be?"

3.1	Photo GPS consistency (not accuracy)	Per-unit centroid clustering · pattern beats accuracy	
3.2	Video site presence	Dickson's L0a + multi-point GPS sampling start/middle/end	SHIPPED
3.3	Video-photo GPS cross-reference	Dickson's L4 + temporal alignment + walk-path proximity	SHIPPED
3.4	Pre-construction baseline + drone reference	Smartphone 360° + drone (revived per Adrian — big devs already drone)	REVIVED
3.5	Project-level geo-tag (Pak Jun #11)	Bank inputs full address; system geocodes — respects Indonesian app convention	
3.6	Satellite imagery cross-check	Sentinel-2 free tier default · Planet Labs premium tier	NEW
3.7	Siteplan-relative cross-reference	Pattern-matching against siteplan — Adrian's archipelago insight	NEW

Time

"Was this captured now, not last week or recycled?"

4.1	Audio fingerprint	Dickson's L0b · Shazam-style spectral upgrade · best singing pitch sentence	SHIPPED
4.2	Photo freshness + monotonic	Dickson's L5 · multi-modal time triangulation upgrade	SHIPPED
4.3	Sun-angle / shadow consistency	Astronomical math — 'we use physics to verify time' · pitch theater	
4.4	Behavioral baseline per Pelaksana	Time-of-day · pace · linguistic · geographic · multi-factor composite	
4.5	External time anchor / public sealing	RFC 3161 TSA preferred over blockchain framing	NEW
4.6	Indonesian-context cross-correlation	BMKG weather · call-to-prayer · uniquely Indonesian moat	NEW

Chain of Custody

"Once evidence enters AutoKon, can it be tampered with?"

5.1	Multi-step approval workflow	Pengawas → SM → PM · quorum-based for high-value · AI-assisted queue	SHIPPED
5.2	Tamper-evident hash chain	Per-project Merkle tree · WORM storage = Phase-0 win, sh a day	NEW
5.3	Periodic public sealing	RFC 3161 TSA · independent third-party (Privy/Tilaka) · transparency log	NEW
5.4	SM digital sign-off / witness	Dickson's exact ask — 'Saksi: Anda lihat hari ini? Ya/Tidak' · 10-min ship	
5.5	Cryptographic per-signer signing	PKI public registry · independently verifiable signatures	NEW
5.6	Immutable audit log	Postgres temporal tables · WAL export to immutable stora	NEW

Independent Verification

"Do you ever rely on something other than your own software?"

6.1	KJPP physical inspection	AutoKon-routed triggers using Dickson's tier classification — flagship pitch line	
6.2	RT/RW or Lurah witness signature	Uniquely Indonesian — no US/EU competitor would think of	REVISED
6.3	Satellite imagery cross-check	Cross-listed with Pillar 3 · 'EU Space Agency Sentinel-2'	
6.4	Drone overflight	Cross-listed with Pillar 3 · developer-supplied	
6.5	Independent third-party signing	Privy/eMeterai · same e-sign as bank's KYC team	NEW

Threat motivation × tech-savviness — the 2x2 that organizes priority.

Most Indonesian fraud is low-tech with high motivation, not the cybersecurity-textbook profile. This shapes ship priority.



Engineered for Indonesian field reality.

No US/EU competitor would think to build any of these. They're the differentiation banks remember.

01

Equatorial sun-angle math

Sun moves $\sim 15^\circ$ /hour at equator vs $\sim 7^\circ$ in mid-latitudes. Indonesia is uniquely sensitive to time-shift.

02

Call-to-prayer (azan) cross-check

If azan is in walk-around audio, cross-reference muezzin schedule for that GPS.

03

BMKG weather correlation

Cross-check claimed weather against Indonesian Meteorological Agency feed.

04

RT/RW witness signatures

Local Indonesian governance officials witness milestones — community-rooted independence.

05

Low-tech consumer-app fraud

Catch Word export, IG Story, screenshots — what Indonesian fraudsters actually do.

06

Siteplan-relative GPS pattern

Pattern beats accuracy — designed for archipelago-reality GPS infrastructure.

Ship in days. Highest leverage.

Items that defend against the high-motivation high-volume Indonesian fraud profile, with zero or near-zero engineering cost.

1.6 A-D

Low-tech fraud detectors

JPEG-quantization, IG-Story, screenshot, photo-of-screen — catches the dominant Indonesian fraud profile

5.2 D

WORM storage config

Single config change. Ships in a day. 'Even AutoKon admins cannot delete evidence.'

5.4 A

Botpress witness prompt

10-minute flow change. Forces SM legal-grade attestation.

3.7 A+B

Centroid + siteplan pattern

Adrian's pattern-beats-accuracy insight. Fixes adjacent-unit confusion in a way Indonesian GPS can support.

4.1 B

Spectral fingerprint

Shazam-style upgrade to existing audio fingerprint. Defeats re-encoding attacks.

6.1 B

AutoKon-routed KJPP

Uses existing tier classification. 'Software pre-screens, KJPP inspects what software flags.'

Ship in 30 days from first signed pilot.

Premium-tier features the first signed bank pilot funds. Public commitment with audit-grade legal weight.

IDENTITY KTP-bound capture (Pak Jun #10) · KYC partnership (Privy/Tilaka) · WhatsApp-native KYC enrollment · liveness via vendor

PLACE Drone fly-over ingest from developer's existing drone · Planet Labs satellite (premium tier)

TIME RFC 3161 TSA public sealing · BMKG weather correlation · call-to-prayer cross-check · multi-segment audio fingerprint

CUSTODY Per-project Merkle tree · per-unit hash chain · RFC 3161 sealing · PKI public registry · audit log with hash links

INDEPENDENT RT/RW witness signature · Privy/eMeterai third-party signing

Roadmap-named, observed-attack-driven shipping.

Named in the pitch for credibility, but engineered only when real-world fraud data justifies the investment.

PHASE 2

Future-tier features

- C2PA content credentials · waits for Pelaksana phone fleet upgrade ~2027+
- Android Play Integrity · requires custom AutoKon camera app
- Mobile-device-bound keys · custom-app pairing
- HSM keys · operational hardening
- Database WAL export · deeper protection
- Event-sourcing architecture · significant rewrite
- PRNU sensor fingerprint · theoretical for Indonesian field
- Bitcoin/Ethereum sealing · only when audience wants blockchain framing

ROADMAP RESERVE

Defer until observed

- Advanced AI-generation defeats — until real attack observed in pilot data
- VFX video forgery — until observed
- Cross-modality consistent forgery — until observed
- Rooted-device sophisticated bypass — until observed at scale
- *These items live in the pitch as 'we have the architectural framework to add these when threats emerge,' not as committed delivery.*

The actual sentences for the bank pitch deck.

AUTHENTICITY

"We screen every photo against state-of-the-art AI-generation detectors — the same techniques used to catch the JAKI Pemprov DKI Gemini-watermark scandal in 2025."

AUTHENTICITY

"We catch the Microsoft-Word-export trick. We catch the Instagram Story overlay trick. We catch screenshots. We've engineered for what Indonesian fraudsters actually do."

IDENTITY

"Bank-PIC signatures go through Privy / eMeterai — the same e-signature infrastructure your KYC team uses."

PLACE

"We don't claim perfect GPS coordinates — Indonesia's archipelago infrastructure makes that impossible. We claim 1000 units have 1000 distinct GPS signatures matching the siteplan layout."

TIME

"Our audio fingerprint catches recycled videos using the same forensic technique Shazam uses to identify songs."

TIME

"If the call to prayer is audible in the walk-around video, our system verifies it matches the muezzin schedule for that location."

CUSTODY

"All evidence files are stored on Write-Once-Read-Many infrastructure. Even AutoKon admins cannot delete or overwrite evidence."

INDEPENDENT

"AutoKon makes KJPP 10x more focused. Software pre-screens; KJPP inspects what software flags as risky."

Two takeaways for the bank.

01

AutoKon is engineered for Indonesian field reality.

Equatorial sun-angle math · BMKG weather feeds · call-to-prayer audio cross-check · RT/RW witness signatures · low-tech consumer-app fraud detectors · siteplan-relative GPS pattern matching. No US/EU competitor would build any of these.

02

AutoKon makes the inspector 10x more focused.

Software catches ~80% of realistic fraud volume deterministically. The residual 20% is structurally chain-of-custody — KJPP at draw milestones, already part of how Indonesian banks operate. Software pre-screens; KJPP inspects what software flags.

Eight decisions before engineering proceeds.

- 1 Is custom AutoKon camera app on the table for PILOT-30? (Resolves Items 1.5B + 2.5A together)
- 2 KYC partnership selection: Privy vs Tilaka vs Sumsb vs Veriff
- 3 Sentinel-2 free vs Planet Labs paid for Phase 0 satellite imagery
- 4 OpenTimestamps blockchain vs RFC 3161 TSA for public sealing language
- 5 WORM storage migration on Supabase — engineering Phase-0 immediate win
- 6 Drone overflight ingest format standardization with developers
- 7 Tier-3 review queue SLA (Dickson's open question 6)
- 8 RT/RW witness program scope and honoraria budget

Six pillars,
thirty-five items,
one **verifiable** bank pitch.

FULL BRIEF

EVIDENCE_INTEGRITY_RESEAR
CH.md

INTERACTIVE

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